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Annexure3b- Complete filing

INVENTION DISCLOSURE FORM

Details of Invention for better understanding:

1. TITLE: Self-Learning Hospital Queue System Using Patient Stress Signals.

2. INTERNAL INVENTOR(S)/ STUDENT(S): All fields in this column are mandatory to be filled

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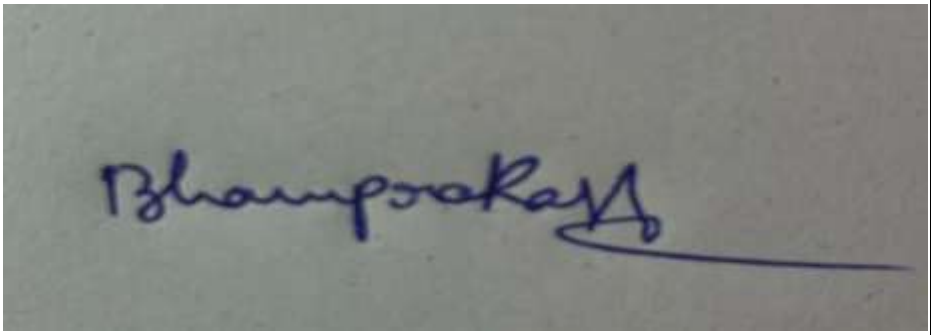
Signature (Mandatory)	

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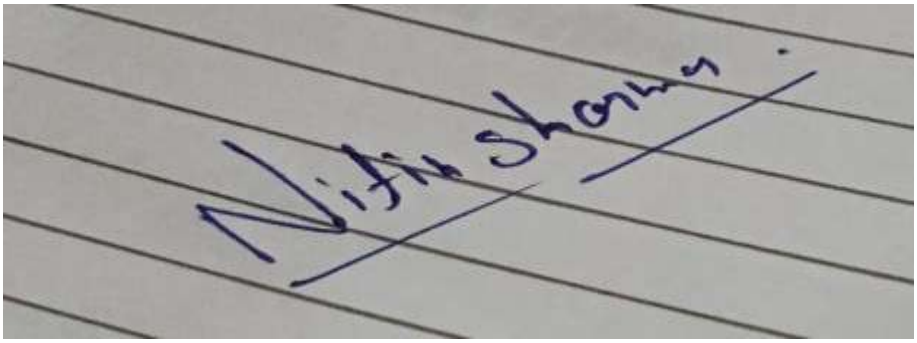
Signature (Mandatory)	

EXTERNAL INVENTOR(S): (INVENTORS NOT WORKING IN LPU)

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***For External Inventors*, NOC (No Objection Certificate) from the affiliated institute/university/Industry/lab etc. is mandatory for each individual inventor and their respective topic. For NOC, format is attached below.**

(FOR ADDITIONAL INVENTORS, PLEASE ADD ROWS)

3. DESCRIPTION OF THE INVENTION:

- **1. Field of the Invention**

The present invention relates generally to hospital management systems and healthcare informatics. More particularly, the invention relates to an artificial intelligence-based self-learning hospital queue management system that dynamically prioritizes patients using real-time physiological and behavioral stress signals.

- **2. Background of the Invention**

Conventional hospital queue systems operate primarily on a first-come, first-served basis or manual triage assessment. In many healthcare facilities, especially high-volume outpatient departments and emergency rooms, such systems may fail to accurately prioritize patients whose medical condition deteriorates while waiting.

Manual triage methods rely heavily on human judgment and periodic monitoring, which may lead to delayed identification of high-risk patients. Furthermore, existing digital token systems do not incorporate real-time physiological data or adaptive learning mechanisms to optimize queue prioritization.

There is therefore a need for an intelligent, automated, and adaptive system capable of continuously monitoring patient stress indicators and dynamically adjusting queue priority in real time.

- **3. Summary of the Invention**

The present invention provides a self-learning hospital queue management system that utilizes multimodal patient stress signals to dynamically determine and adjust queue priority levels.

The system integrates physiological sensors, behavioral analysis modules, and machine learning algorithms to compute a real-time stress index for each patient. Based on this computed index, a dynamic priority score is generated, which automatically reorders the patient queue to ensure timely medical attention for high-risk individuals.

The invention further incorporates a continuous learning mechanism that improves prioritization accuracy over time using historical data, treatment outcomes, and clinician feedback.

- **4. Detailed Description of the Invention**
- **4.1 System Architecture**

The system comprises the following primary components:

- **(a) Data Acquisition Module**

This module collects real-time patient data from one or more of the following sources:

- **Wearable medical sensors measuring heart rate, blood pressure, oxygen saturation, and body temperature**
- **Camera systems capturing facial expressions and behavioral indicators**
- **Microphone systems analyzing voice tone and speech stress patterns**
- **Patient self-reported data via a mobile or kiosk interface**
- **Electronic medical record (EMR) integration**

The collected data is transmitted securely to a central processing unit.

- **(b) Stress Signal Processing Module**

The processing module analyzes incoming data using signal processing techniques and machine learning models. The module:

- **Extracts relevant physiological features**
- **Identifies abnormal stress patterns**
- **Applies classification algorithms to determine stress severity**

A composite "Patient Stress Index (PSI)" is calculated using weighted parameters derived from multiple data sources.

- **(c) Dynamic Priority Engine**

The Dynamic Priority Engine converts the Patient Stress Index into a queue priority score.

The priority score is determined using:

- **Current stress level**
- **Historical medical records**
- **Pre-existing conditions**
- **Age and risk category**
- **Duration of waiting time**

Based on this computed score, the system automatically reorders the hospital queue in real time.

- **(d) Self-Learning Mechanism**

The system incorporates a self-learning artificial intelligence model that continuously improves performance through:

- **Reinforcement learning based on patient outcomes**
- **Feedback from medical professionals**
- **Historical queue data analysis**

The model adapts weight parameters and decision thresholds over time to enhance accuracy and reduce false prioritization.

- **(e) Alert and Escalation Module**

If the Patient Stress Index crosses a predefined threshold, the system:

- **Sends immediate alerts to medical staff**
- **Flags the patient as high priority**
- **Triggers emergency intervention protocols when necessary**

- **5. Unique Features of the Invention**

The invention provides several novel and distinctive features:

1. **Real-time multimodal stress detection integrated with queue management**
2. **Automated dynamic queue reordering without manual intervention**
3. **Self-learning adaptive prioritization model**
4. **Continuous monitoring during waiting period**
5. **Integration capability with existing hospital information systems**
6. **Secure handling of sensitive patient data**

- **6. Advantages of the Invention**

The system offers the following advantages:

- **Early detection of medical deterioration**
- **Reduced patient risk during waiting periods**
- **Improved efficiency of hospital workflow**
- **Objective and data-driven prioritization**
- **Reduced dependency on manual triage alone**

A. **PROBLEM ADDRESSED BY THE INVENTION:** Existing hospital queue management systems primarily operate on a first-come, first-served basis or rely on one-time manual triage assessments. These systems do not continuously monitor patients after registration, nor do they dynamically adjust queue priority based on changes in a patient's physiological or psychological condition while waiting.

B. As a result, patients experiencing increasing stress, medical instability, or sudden deterioration may not receive timely attention, potentially leading to adverse health outcomes. Therefore, there is a need for an intelligent and adaptive queue system capable of real-time stress monitoring and automatic priority adjustment to enhance patient safety and hospital efficiency.

C. Additionally, current systems lack a self-learning mechanism that can analyze historical patient data and outcomes to improve future prioritization decisions. Therefore, there is a need for an intelligent and adaptive queue system capable of real-time stress monitoring, automatic priority adjustment, and continuous learning to enhance patient safety and hospital efficiency.

- **OBJECTIVE OF THE INVENTION :**
- To develop an intelligent hospital queue management system capable of continuously monitoring patient physiological and behavioral stress signals during the waiting period.
- To provide a dynamic priority allocation mechanism that automatically reorders patient queues in real time based on computed stress levels and medical risk assessment.
- To incorporate a self-learning algorithm that improves prioritization accuracy over time using historical data, patient outcomes, and clinical feedback.
- To enhance patient safety and reduce medical risk by enabling early detection of condition deterioration and timely escalation to medical staff.

C. STATE OF THE ART/ RESEARCH GAP/NOVELTY: Hospital queue management and patient prioritization have been active areas of research in healthcare informatics and medical systems engineering. Traditional approaches to patient prioritization primarily rely on manual triage systems, where healthcare personnel assess the severity of a patient's condition during initial registration and assign priority categories accordingly. Such systems, while widely used, depend on human judgment and periodic reassessment, and do not provide continuous monitoring of patient condition during waiting periods.

Some existing digital queue management systems improve upon token-based methods by enabling appointment scheduling and estimated wait time notifications. However, these systems still follow static priority rules and do not incorporate physiological data to dynamically adjust queues in real time.

Advances in wearable sensor technologies and biomedical signal processing have enabled continuous monitoring of vital parameters such as heart rate, blood pressure, oxygen saturation, and respiratory patterns. Research in healthcare sensing has demonstrated the feasibility of using such data for early detection of clinical deterioration in hospitalized patients. Similarly, affective computing research has explored the use of facial expression analysis, speech pattern recognition, and behavioral indicators to estimate emotional stress and discomfort in individuals.

Despite these advances in sensing and stress detection, current systems typically focus on in-patient monitoring or mental health applications and are not integrated with hospital queue management frameworks. Some academic studies and experimental systems have explored the use of artificial intelligence for clinical decision support and triage assistance at the point of care, but these systems do not continuously monitor patients while waiting or adjust queue priority dynamically based on multimodal stress indicators.

Furthermore, existing prioritization algorithms generally lack adaptive or self-learning mechanisms that can improve performance over time based on historical outcomes, clinician feedback, or evolving patient patterns. Machine learning models used in healthcare tend to be trained offline for specific tasks such as disease prediction, risk stratification, or diagnostic classification, rather than real-time queue adjustment based on continuously streaming physiological and behavioral signals.

Consequently, there remains a significant gap in the development of an integrated system that combines real-time stress signal acquisition, adaptive machine learning, and dynamic queue management within hospital settings. The present invention addresses this gap by providing a self-learning hospital queue management system capable of continuously monitoring patient stress indicators, generating real-time priority scores, and dynamically reordering queues to ensure timely medical attention for patients at higher risk of deterioration.

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Sr. No.	Patent I'd	Abstract	Research Gap	Novelty
1.				
2.				
3.				

D. DETAILED DESCRIPTION: in a patent provides a complete explanation of the invention's technical details, allowing patent examiners and people interested in the idea to have an accurate understanding of its design and functionality?

E. RESULTS AND ADVANTAGES: Share the results and advantages and superiority over the existing prior art.

F. EXPANSION: Any variables which are necessary for your invention to be covered?

G. WORKING PROTOTYPE/ FORMULATION/ DESIGN/COMPOSITION: Is your working prototype or other ready. Provide the images/data of the prototype. If no, how much time is required for the same?

G. EXISTING DATA: Any clinical or comparative data necessary enough to support your invention. (Comparative)

4. USE AND DISCLOSURE (IMPORTANT): Please answer the following questions:

A. Have you described or shown your invention/ design to anyone or in any conference?	YES ( )	NO ( )
B. Have you made any attempts to commercialize your invention (for example, have you approached any companies about purchasing or manufacturing your invention)?	YES ( )	NO ( )

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C. Has your invention been described in any printed publication, or any other form of media, such as the Internet?	YES ( )	NO ( )
D. Do you have any collaboration with any other institute or organization on the same? Provide name and other details.	YES ( )	NO ( )
E. Name of Regulatory body or any other approvals if required.	YES ( )	NO ( )

5. Provide links and dates for such actions if the information has been made public (Google, research papers, YouTube videos, etc.) before sharing with us.

6. Provide the terms and conditions of the MOU also if the work is done in collaboration within or outside university (Any Industry, other Universities, or any other entity).

7. Potential Chances of Commercialization.

8. List of companies which can be contacted for commercialization along with the website link.

9. Any basic patent which has been used and we need to pay royalty to them.

10**. FILING OPTIONS:** Please indicate the level of your work which can be considered for provisional/ complete/ PCT filings (Mandatory to mention).

11. **KEYWORDS:** Please provide right keywords for searching your invention.

NO OBJECTION CERTIFICATE

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This is to certify that University/Organization Name or its associates shall have no objection if Lovely Professional University files an IPR (Patent/Copyright/Design/any other…….) entitled "……." including the name(s) of,.....…as inventors who is(are) student(s)/employee(s) studying/ working in our University/ organization.

Further Name of the University/Organization shall not provide any financial assistance in respect of said IPR nor shall raise any objection later with respect to filing or commercialization of the said IPR or otherwise claim any right to the patent/invention at any stage.

(Authorised Signatory)